Experiment 1

Aim : NumPy, Pandas, Matplotlib & Seaborn for Machine Learning

Code :

#Load Dataset :

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv('student_performance.csv')
```

 # Exercise 1: NumPy Basics

```
print("--- Exercise 1: NumPy Basics ---")

# 1. Load Final_Score as NumPy array
final_scores = df['Final_Score'].to_numpy()

# 2. Compute mean, median, and standard deviation
mean_val = np.mean(final_scores)
median_val = np.median(final_scores)
std_val = np.std(final_scores)

print(f"Mean Final Score: {mean_val:.2f}")
print(f"Median Final Score: {median_val:.2f}")
print(f"Standard Deviation: {std_val:.2f}")

# 3. Perform Min-Max normalization
# Formula: (x - min) / (max - min)
min_val = np.min(final_scores)
max_val = np.max(final_scores)
normalized_scores = (final_scores - min_val) / (max_val - min_val)

print("Normalized Scores (first 5):", np.round(normalized_scores[:5], 2))
print("\n")
```

 # Exercise 2: Pandas Data Handling

```
print("--- Exercise 2: Pandas Data Handling ---")
```

```
# 1. Load CSV (Already done above)

# 2. Check shape, columns, and missing values
print(f"Shape of dataset: {df.shape}")
print(f"Columns: {df.columns.tolist()}")
print("Missing Values:\n", df.isnull().sum())

# 3. Create Performance label based on Final_Score
# Criteria: >= 75 (Good), >= 60 (Average), < 60 (Low)
def classify_performance(score):
    if score >= 75:
        return 'Good'
    elif score >= 60:
        return 'Average'
    else:
        return 'Low'

df['Performance'] = df['Final_Score'].apply(classify_performance)
print("\nData with Performance Label (first 5 rows):")
print(df[['Final_Score', 'Performance']].head())
print("\n")
```

 # Exercise 3: Matplotlib Visualization

```
print("--- Exercise 3: Matplotlib Visualization ---")

# 1. Line plot: Hours_Studied vs Final_Score
# We sort by Hours_Studied to make the line plot look clean
df_sorted = df.sort_values(by='Hours_Studied')

plt.figure(figsize=(8, 5))
plt.plot(df_sorted['Hours_Studied'], df_sorted['Final_Score'], marker='o', linestyle='-', color='blue')
plt.title('Line Plot: Hours Studied vs Final Score')
plt.xlabel('Hours Studied')
plt.ylabel('Final Score')
plt.grid(True)
plt.show()

# 2. Histogram of Final_Score
plt.figure(figsize=(8, 5))
plt.hist(df['Final_Score'], bins=10, color='skyblue', edgecolor='black')
plt.title('Histogram of Final Score')
plt.xlabel('Final Score')
plt.ylabel('Frequency')
```

```
plt.show()


# Exercise 4: Seaborn Visualization

print("--- Exercise 4: Seaborn Visualization ---")

# 1. Scatter plot using seaborn
plt.figure(figsize=(8, 5))
sns.scatterplot(data=df, x='Hours_Studied', y='Final_Score', hue='Performance', palette='viridis',
s=100)
plt.title('Seaborn Scatter Plot: Hours vs Score')
plt.show()

# 2. Heatmap for correlation analysis
plt.figure(figsize=(8, 6))
# Select only numeric columns for correlation matrix
numeric_df = df.select_dtypes(include=[np.number])
corr_matrix = numeric_df.corr()
sns.heatmap(corr_matrix, annot=True, cmap='coolwarm', fmt=".2f")
plt.title('Correlation Heatmap')
plt.show()

# 3. Boxplot for categorical analysis
plt.figure(figsize=(8, 5))
sns.boxplot(data=df, x='Performance', y='Final_Score', order=['Low', 'Average', 'Good'],
palette='Set2')
plt.title('Boxplot: Performance vs Final Score')
plt.show()
```

Outputs :

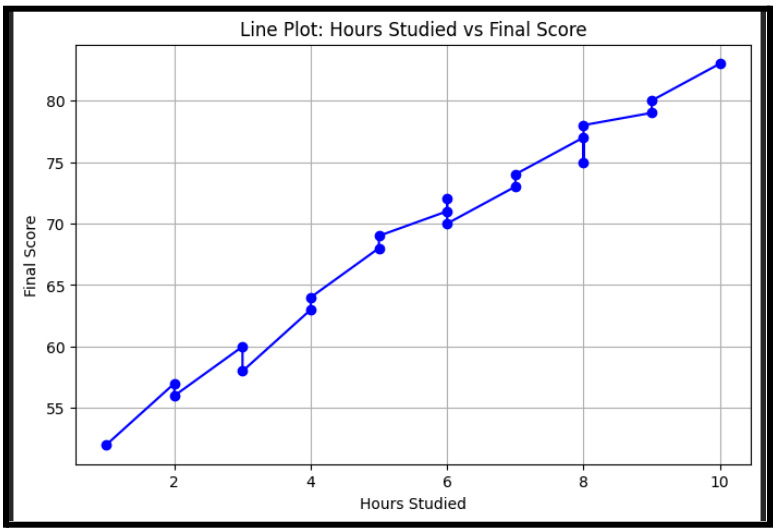
1) Numpy Basic Computations

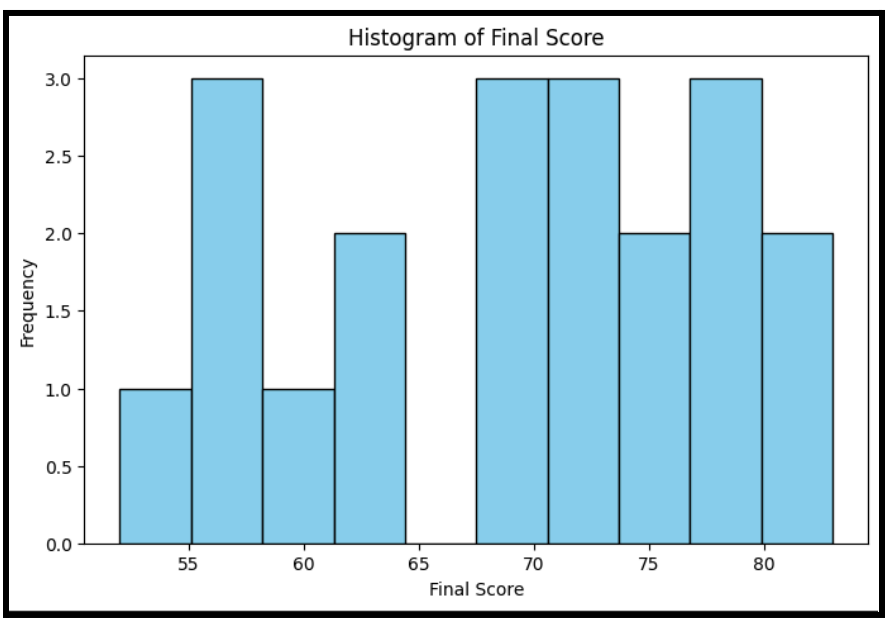
```
--- Exercise 1: NumPy Basics ---
Mean Final Score: 68.95
Median Final Score: 70.50
Standard Deviation: 8.71
Normalized Scores (first 5): [0.   0.16 0.26 0.39 0.52]
```

2) Calculate the Performance Label

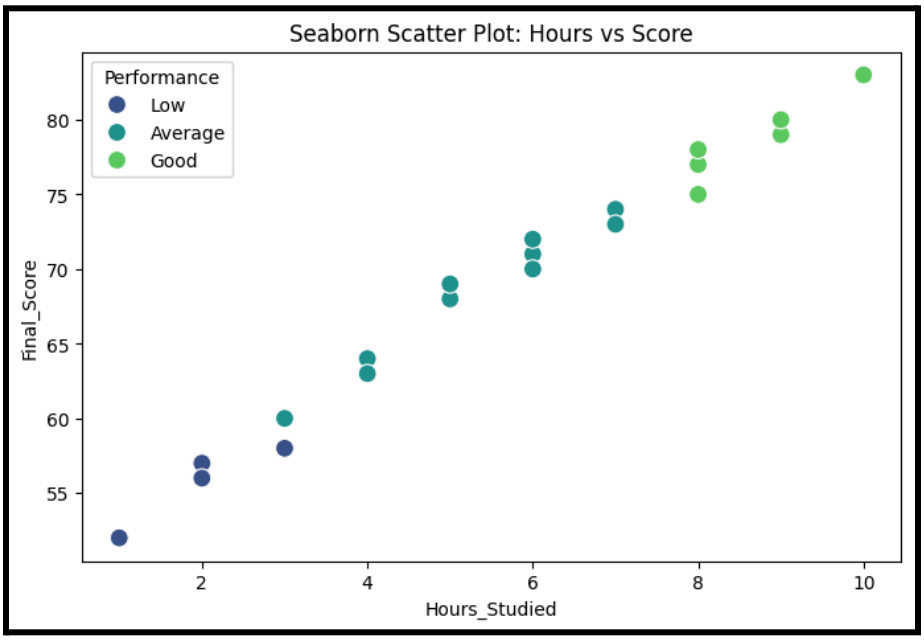
```
--- Exercise 2: Pandas Data Handling ---
Shape of dataset: (20, 5)
Columns: ['Hours_Studied', 'Attendance', 'Assignment_Score', 'Midterm_Score', 'Final_Score']
Missing Values:
 Hours_Studied       0
Attendance          0
Assignment_Score    0
Midterm_Score       0
Final_Score         0
dtype: int64

Data with Performance Label (first 5 rows):
   Final_Score Performance
0           52         Low
1           57         Low
2           60     Average
3           64     Average
4           68     Average
```

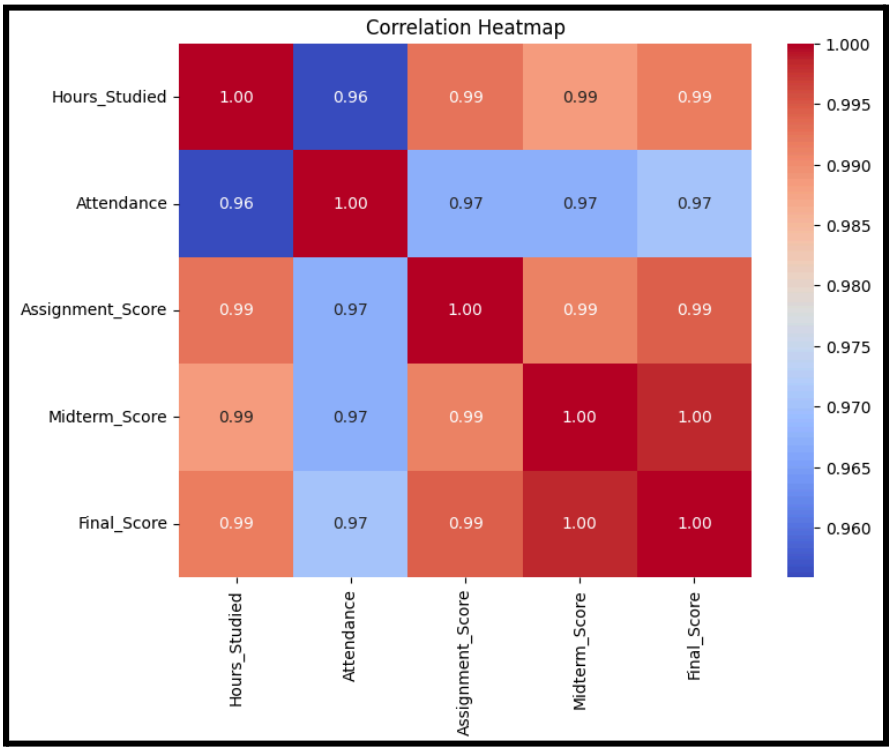
3) Matplotlib Visualization



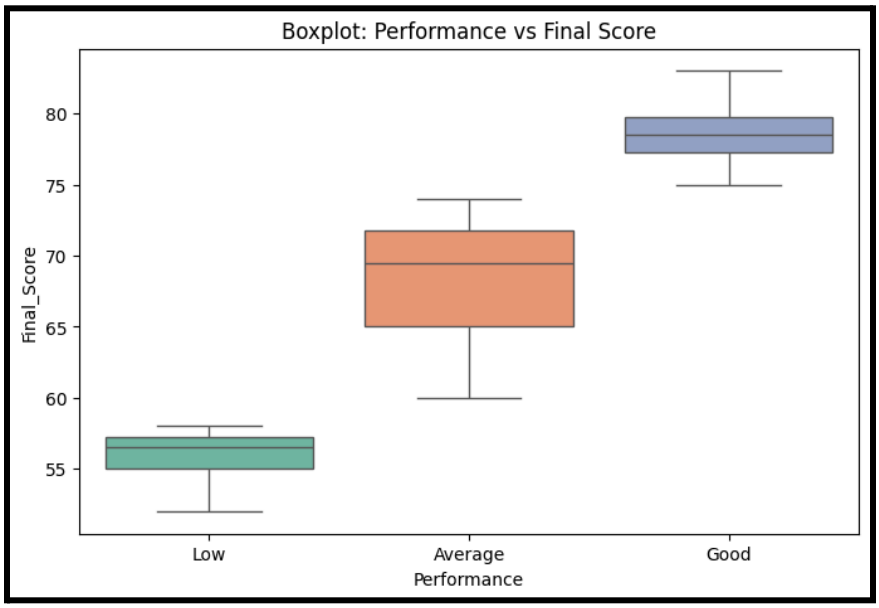


4) Seaborn Visualization





5) Boxplot Visualization



Conclusion : Hence we have successfully applied the NumPy, Pandas, Matplotlib & Seaborn for Machine Learning